Market Regime Detection and Fragility Dynamics

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Introduction

Financial markets evolve through distinct regimes characterized by varying levels of volatility, correlation, and macroeconomic stress.

This study introduces a Market Regime Engine designed to detect transitions between stable and stressed market conditions using a fragility-based framework.

The model integrates:

- Cross-asset volatility dynamics
- Correlation structures
- Foreign exchange risk
- Tail risk measures (VaR and CVaR)

The objective is to identify early signals of regime shifts and systemic fragility. This approach extends previous work [2] by incorporating heavy-tail behavior and network-driven contagion.

Model Framework

Let $r_i(t)$ denote the log-return of asset i.

We define a portfolio return:

$$R_p(t) = \frac{1}{N} \sum_{i=1}^{N} r_i(t)$$

The system incorporates multiple risk dimensions derived from rolling windows:

- Volatility: $\sigma(t)$
- Correlation: $\rho(t)$

- FX volatility: $\sigma_{FX}(t)$

Tail Risk Measures

Value-at-Risk (VaR):

$$VaR = \inf\{x : P(R_p \leq x) \geq 1 - \alpha\}$$

Conditional Value-at-Risk (CVaR):

$$CVaR = E[R_p \mid R_p \leq VaR]$$

These measures capture extreme downside risk and are used to monitor stress accumulation.

Fragility Index

Systemic fragility is defined as a weighted combination of normalized risk components:

$$F(t) = w_1 \cdot Z_\sigma(t) + w_2 \cdot Z_\rho(t) + w_3 \cdot Z_{FX}(t)$$

where:

- $Z_\sigma(t)$ is volatility (robust-scaled)
- $Z_\rho(t)$ is average correlation
- $Z_{FX}(t)$ is FX volatility

Weights satisfy:

$$w_1 + w_2 + w_3 = 1$$

In this model:

$$(w_1, w_2, w_3) = (0.45, 0.35, 0.20)$$

Normalized Fragility Score

To map fragility into an interpretable scale, we apply a logistic transformation:

$$Fragility(t) = \frac{100}{1 + e^{-kF(t)}}$$

where k controls sensitivity.

The index is bounded in $[0, 100]$, facilitating regime classification.

Market Regime Classification

Market regimes are defined as:

- **Risk-On:** $Fragility < 35$
- **Neutral:** $35 \leq Fragility < 65$
- **Risk-Off:** $Fragility \geq 65$

These regimes capture transitions between stable and stressed market environments.

Correlation Network

A dynamic network is constructed using rolling correlations:

$$\rho_{ij}(t) = \mathrm{Corr}(r_i, r_j)$$

Edges represent strong dependencies, and node centrality reflects systemic importance.

Highly correlated nodes amplify shock transmission and contribute to systemic fragility.

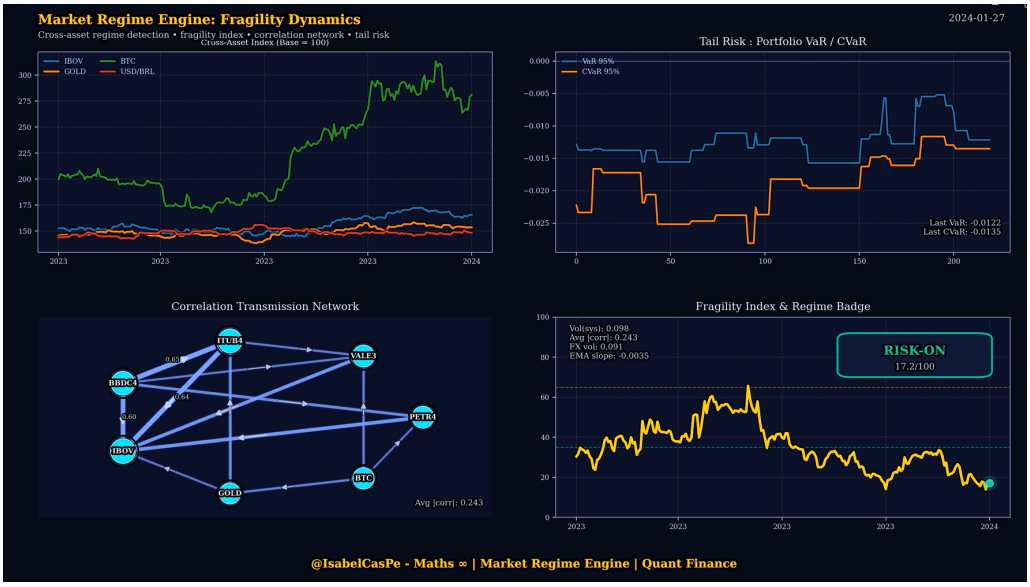

Figure 1: **Market regime dynamics and fragility evolution (V. 13).** The visualization integrates cross-asset price dynamics, tail risk measures, correlation networks, and the Fragility Index. The regime badge highlights transitions between risk-on, neutral, and risk-off states, revealing how volatility, correlation, and FX stress jointly drive systemic conditions.

Results and Interpretation

The V.13 framework reveals that market regimes evolve continuously rather than discretely.

Key observations include:

- Fragility increases with correlation clustering
- FX volatility acts as an external stress amplifier
- Tail risk precedes regime transitions

The model captures gradual transitions from stable to stressed states, providing early insight into systemic instability.

Conclusion

The Market Regime Engine provides a unified framework for detecting systemic conditions in financial markets.

By integrating volatility, correlation, and FX risk into a single fragility measure, the model enables:

- Early detection of regime shifts
- Quantification of systemic stress
- Improved interpretation of market dynamics

This approach extends traditional risk models by incorporating structural and cross-asset dependencies.

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